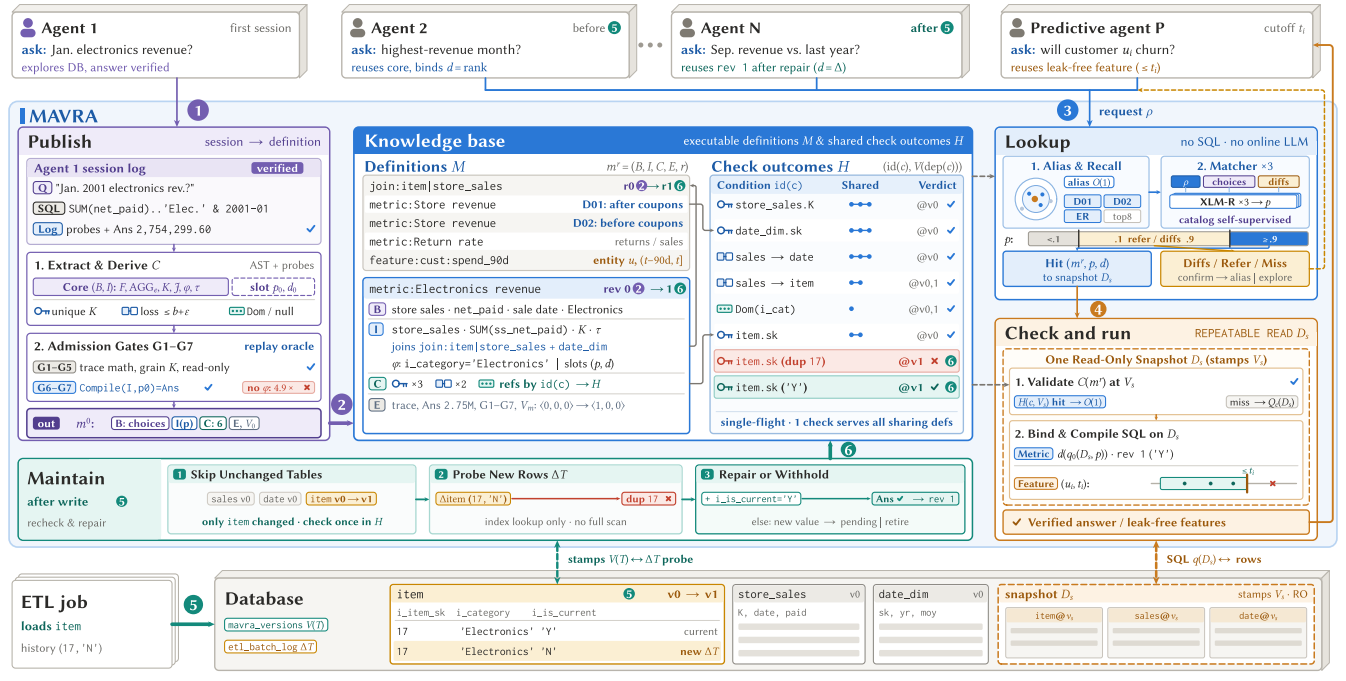

Agent 1
first session
ask: Jan. electronics revenue?
explores DB, answer verified
Agent 2
before 5
ask: highest-revenue month?
reuses core, binds d = rank
Agent N
after 5
ask: Sep. revenue vs. last year?
reuses rev 1 after repair (d = Δ)
Predictive agent P
cutoff t_i
ask: will customer u_i churn?
reuses leak-free feature (≤ t_i)
MAVRA
1
Publish
session → definition
Agent 1 session log
verified
Q "Jan. 2001 electronics rev.?"
SQL SUM(net_paid)..'Elec.' & 2001-01
Log probes + Ans 2,754,299.60
1. Extract & Derive C
AST + probes
Core (B, I): F, AGG_a, K, J, φ, τ
slot p_0, p_1
unique K
loss ≤ b+ε
Dom / null
2. Admission Gates G1–G7
replay oracle
G1–G5 trace math, grain K, read-only
G6–G7 Compile(I,p0)=Ans
no φ: 4.9×
out m^0: B: choices I(p) C: 6 E, V_0
2
Knowledge base
executable definitions M & shared check outcomes H
Definitions M
m^r = (B, I, C, E, r)
join:item|store_sales
r0 2 → r1 6
metric:Store revenue
D01: after coupons
metric:Store revenue
D02: before coupons
metric:Return rate
returns / sales
feature:cust:spend_90d
entity u_i (t−90d, t]
metric:Electronics revenue
rev 0 2 → 1 6
B store sales · net_paid · sale date · Electronics
I store_sales · SUM(ss_net_paid) · K · τ
joins join:item|store_sales + date_dim
φ: i_category='Electronics' | slots (p, d)
C ×3 ×2 refs by id(c) → H
E trace, Ans 2.75M, G1–G7, V_m: ⟨0, 0, 0⟩ → ⟨1, 0, 0⟩
Check outcomes H
(id(c), V(dep(c)))
Condition id(c)
Shared
Verdict
store_sales.K
@v0
date_dim.sk
@v0
sales → date
@v0
sales → item
@v0,1
Dom(i_cat)
@v0,1
item.sk
@v0
item.sk (dup 17)
@v1
6
item.sk ('Y')
@v1
6
single-flight · 1 check serves all sharing defs
3
request ρ
Lookup
no SQL · no online LLM
1. Alias & Recall
alias O(1)
D01
D02
ER
top8
2. Matcher ×3
ρ
choices
diffs
XLM-R ×3 → p
catalog self-supervised
p:
<.1
.1 refer / diffs .9
≥.9
Hit (m^r, p, d)
to snapshot D_s
Diffs / Refer / Miss
confirm → alias | explore
4
Check and run
REPEATABLE READ D_s
One Read-Only Snapshot D_s (stamps V_s)
1. Validate C(m^r) at V_s
H(c, V_s) hit → O(1)
miss → Q_c(D_s)
2. Bind & Compile SQL on D_s
Metric d(q_r(D_s, ρ)) · rev 1 ('Y')
Feature ⟨u_i, t_i⟩:
≤ t_i
Verified answer / leak-free features
6
Maintain
after write 5
recheck & repair
1 Skip Unchanged Tables
sales v0
date v0
item v0 → v1
only item changed · check once in H
2 Probe New Rows ΔT
Δitem ('Y', 'N')
dup 17
index lookup only · no full scan
3 Repair or Withheld
+ i_is_current='Y'
Ans ✓ → rev 1
else: new value → pending | retire
stamps V(T) ↔ ΔT probe
SQL q(D_s) ↔ rows
ETL job
loads item
history ('Y', 'N')
5
Database
mavra_versions V(T)
etl_batch_log ΔT
item
5
v0 → v1
i_item_sk
i_category
i_is_current
17
'Electronics'
'Y'
current
17
'Electronics'
'N'
new ΔT
store_sales v0
K, date, paid
date_dim v0
sk, yr, moy
snapshot D_s
stamps V_s · RO
item@v_s
sales@v_s
date@v_s